Bring My Cup!

Personalizing Vision-Language-Action Models with Visual Attentive Prompting

Sangoh Lee, Sangwoo Mo†, Wook-Shin Han†

†: Equal advising

Check Our Project Page!

Check My Profile!

ICML International Conference On Machine Learning

POSTECH

'Cup' is easy. 'My Cup" is hard.



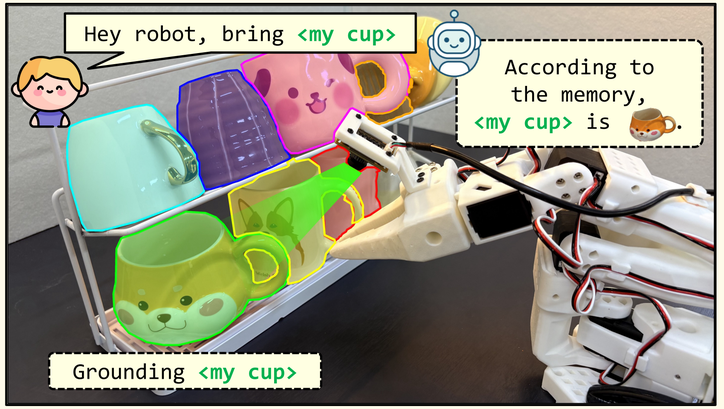
VAP makes 'My' visible for frozen VLAs
Show it → Find it → Mark it → Move it

1. Show user's cup

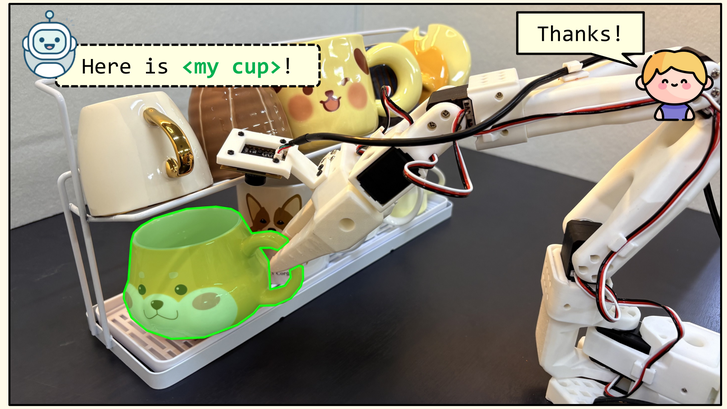


5 images of <my cup>

2. Find and mark user's cup using memory

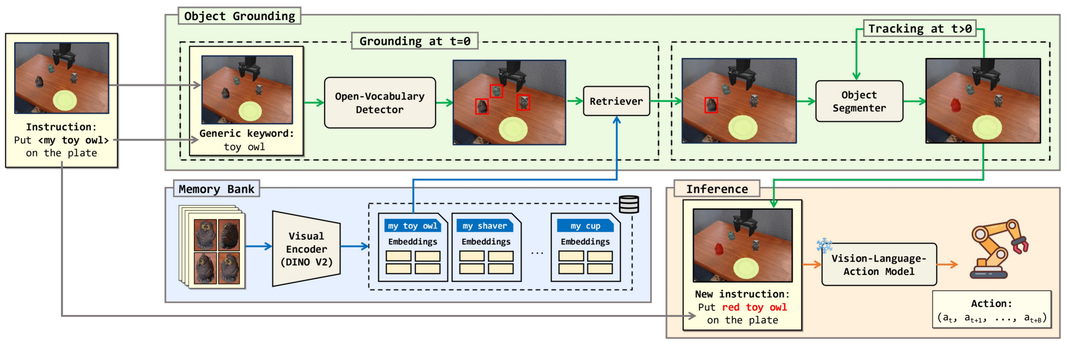


3. Bring the user's cup!

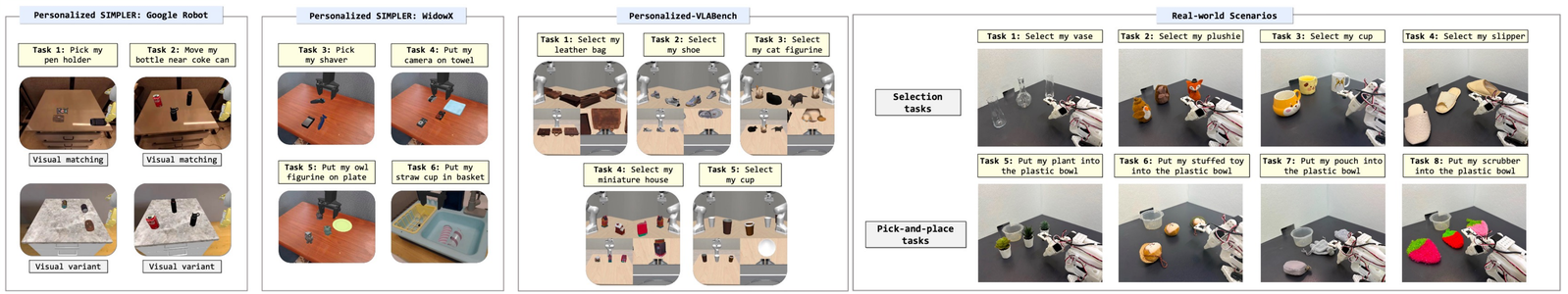


Overview of VAP

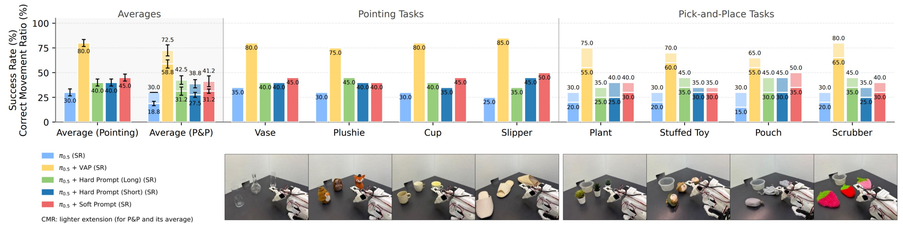
- **VLAs fail at personalized commands** like "my cup."
- Personalization requires **distinguishing a specific object** from look-alike distractors.
- We propose **VAP**, a training-free visual grounding and prompting framework to solve this.



Benchmarks



Quantitative Results



30.0 → 80.0%
Success rate, on selection

18.8 → 58.8%
Success rate, on PnP task

+ 0.02 s / step
Little tracking overhead

Detailed Results

Method	Task 1: Pick my pen holder		Task 2: Move my bottle near coke can	
	CMR (%)	SR (%)	CMR (%)	SR (%)
Track 1: Visual Matching with Unseen Personalization Objects				
π_0	10.5	8.5	50.4	48.6
π_0 + Soft Prompt	33.1	24.0	60.0	51.7
π_0 + Hard Prompt (Short)	11.0	8.9	66.7	57.5
π_0 + Hard Prompt (Long)	11.2	9.1	64.5	58.8
π_0 + VAP	**89.2**	**60.3**	**83.2**	**75.0**
Track 2: Variant Aggregation with Unseen Personalization Objects				
π_0	17.4	10.2	49.2	41.3
π_0 + Soft Prompt	31.8	26.4	57.7	44.7
π_0 + Hard Prompt (Short)	21.6	14.4	71.5	54.2
π_0 + Hard Prompt (Long)	17.3	12.6	67.7	51.3
π_0 + VAP	**87.3**	**58.2**	**72.4**	**62.5**

Method	Task 3: Pick shaver		Task 4: Put camera		Task 5: Put owl		Task 6: Put straw cup	
	CMR (%)	SR (%)	CMR (%)	SR (%)	CMR (%)	SR (%)	CMR (%)	SR (%)
π_0	0.0	0.0	34.2	31.7	35.2	30.3	80.6	27.8
π_0 + Soft Prompt	25.0	8.3	54.2	41.7	79.2	62.5	87.5	29.2
π_0 + Hard Prompt (Short)	0.0	0.0	50.0	25.0	37.5	33.3	83.3	29.2
π_0 + Hard Prompt (Long)	0.0	0.0	53.2	27.9	36.0	31.6	83.4	30.1
π_0 + VAP	**82.9**	**71.3**	**92.1**	**92.1**	**100.0**	**95.0**	**100.0**	**75.6**

Method	Task 1: Leather bag	Task 2: Shoe	Task 3: Cat figurine	Task 4: Miniature house	Task 5: Cup
$\pi_{0.5}$	33.6	30.4	35.2	29.6	40.4
$\pi_{0.5}$ + Soft Prompt	51.2	40.8	44.4	39.6	54.8
$\pi_{0.5}$ + Hard Prompt (Short)	52.4	37.6	44.0	38.8	51.6
$\pi_{0.5}$ + Hard Prompt (Long)	52.0	39.6	42.8	39.6	50.0
$\pi_{0.5}$ + VAP	**89.2**	**54.0**	**51.6**	**52.4**	**60.8**